| UCE605: TRANSPORTATION ENGINEERING-II | | | | |
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| | **3** | **1** | **0** | **3.5** |

**Course Objective**: This subject aims to develop an understanding of the structural & geometrical design of various components of railway track as per the Indian railways guidelines. Further to this, subject also aims at introducing the detail concepts of the airport engineering and to give the students the confidence of delivering a complete geometric and structural design of runway, taxiway and apron pavements

**Railway Engineering**:

**Permanent way specifications**: Gauges in railway tracks, typical railway track cross-section, coning of wheels

**Rails:** Function of rails, requirement of rails, types of rail sections – comparison of rail types, length of rail, rail wear, rail failures, creep of rails, rail fixtures and fastenings – Fish plates, spikes, bolts, chairs, keys, bearing plates

**Sleepers**: Functions and requirements of sleepers, classification of sleepers, timber, metal and concrete sleeper, comparison of different types of sleepers, spacing of sleepers and sleeper density

**Ballast**: Function and requirements of ballast, types, comparison of ballast materials

**Geometric design**: Alignment design, horizontal curves, super elevation, equilibrium, cant and cant deficiency, length & setting out of transition curve, gradients and grade compensation, negative super elevation design

**Points and crossings**: Introduction, necessity of points and crossings, design of a turnout as per Indian railways specifications

**Signaling and interlocking:** Objects of signaling, engineering principle of signaling, classification of signaling, control of train movements, interlocking definition, necessity and function of interlocking, methods of interlocking, mechanical devices for inter locking, Traction and tractive resistances, stresses in track, Hauling capacity of locomotive, modernization of railway track

**Airport Engineering**:

**Airport Planning:** Airport site selection, various surveys for site selection. Classifications of obstructions, Imaginary surfaces, Approach zone and turning zone, Runway orientation using wind rose diagrams, basic runway length, corrections for elevation, temperature & gradient, airport classification

**Runway & Taxiway Design**: Geometric design of runway as per ICAO & FAA guidelines, taxiway layout, geometric design standards for taxiway & aprons, Rapid Exit Taxiways, Structural design of runway pavements, Design of flexible and rigid runways as per FAA

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| procedure using FAARFIELD and PCA method, Design of joints for airport pavements, Specifications for the different layers of runway and taxiway pavements, Pavement Evaluation for runway & taxiway, LCN-PCN method, design of overlay using COMFAA & ELMOD, Pavement management systems for runway pavements<br>**Airport Layouts:** Terminal area, parking area, apron & hanger typical airport layouts, Lightings and markings design for airside area of an airport |
| **Experimental Project/assignment/Micro Project: **<br>1**.** To analyze & design the flexible & rigid airport pavements using FAARFIELD.<br>2. To design the turnout as per the Indian Railway specifications.<br>3. To perform the data analysis for developing management systems for airport pavements. |
| **Course Learning Objectives (CLO)**<br>Upon the completion of this course, the students will be able to:<br>1. Determine the runway orientation and the runway length as per FAA &ICAO guidelines.<br>2. Design the airport pavements including air-side marking & lighting as per ICAO & FAA guidelines<br>3. Evaluate pavement and learn the concept of pavement maintenance management system.<br>4. Employ Railway Track specifications and perform geometric design of the railway track.<br>5. Design turnout and crossings as per the Indian Railways |
| **Text Books:**<br>1. Arora and Saxena, Railway Engineering, Dhanpat Rai & Sons, New Delhi (2006)<br>2. Khanna, Arora & Jain, Airport Planning and Design, Nem Chand & Brothers, Roorkee (1999)<br>3. ICAO and FAA, various advisory circulars guidelines (2018)<br><br>**Reference Books:**<br>1. Rangawala, Railway Engineering, Charotar Publishing House, Anan (2017)<br>2. Aggarwal M.M., and Satish Chandra Railway Engineering, Oxford University Press (2013)<br>3. Horenjeff Robert, Airport Engineering, McGraw Hill International Publisher (2010) |

**Evaluation Scheme:**

| Sr. No. | Evaluation Elements | Weights (%) |
|---|---|---|
| 1. | MST | 30 |
| 2. | EST | 40 |
| 3. | Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations) | 30 |